

Air Traffic Control - The eye in the sky

Surely you haven't given a thought to how a plane takes off, flies and lands without tripping over things, even while you were jet setting off to some destination! Well, the entire process is controlled by what is known as the Air Traffic Control (ATC) system.



TRACON - ATC RADAR

Terminal Radar Approach Control or TRACON is a facility that tracks the plane up to about 80km from the airport, guiding it during takeoff and landing. TRACONS are intermediaries of an aircraft's radio contact to and from the Air Route Traffic Control Centers that guides an aircraft between two airports -- these are strategically placed air space monitoring systems. Along with normal radio communication, a TRACON situated in an airport's ATC also utilizes good old human techniques of visually identifying a plane as it approaches an airport. The following are the three stages of an aircraft as it lands at an airport.



EN-ROUTE

One, or multiple ARTCC (Air Route Traffic Control Center) will monitor and guide the plane till it reaches within 80km of the destination airport, at which point the control is transferred to local ATC tower at the airport.



APPROACH

TRACON takes charge of monitoring the plane as it approaches the airport at this stage. If required, the plane is asked to circle the airport for a landing slot -- this happens whenever too many aircrafts land or takeoff simultaneously.



LANDING

Before the plane lands, the ground controller checks visually (with binoculars) and via the ground radars and identifies the perfect window for the aircraft to land. If all's clear, the green signal is lit.